Multiple solutions for the decomposition of simple simplex systems

Benjamin Evans

Department of Linguistics,
Institute for Advanced Computational Sciences,
Stony Brook University, NY USA

Rutgers Subregular Morphophonology Workshop '26

Nov 07 2026

Pitch

Existing decomposition frameworks gauge success as a measure of composite exactness.[†]

Pitch

Existing decomposition frameworks gauge success as a measure of composite exactness.[†]

Pitch

We should be interested in meaningfully distinct solutions in similar complexity classes that surface (via composition) identically to the generating object.

Outline

Core framework

Simple simplex systems
- $\mathbb{H}, \mathbb{R}$-learning perspective
- Conditions on interaction

Multiple solutions
- $\mathbb{H}^2, \mathbb{R}^2$ decomposition framework
- Applying the $\mathbb{H}^2, \mathbb{R}^2$ framework
- Templates
- Hypothesis evaluation
- Path trees

Appendix
- Conditions on interaction

Core framework

Core framework: $\mathbb{H}, \mathbb{R}$ learning

Identification in the limit from positive data (Gold 1978; Higuera 1997)

Core framework: $\mathbb{H}, \mathbb{R}$ learning

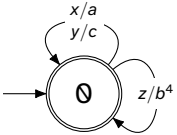
Identification in the limit from positive data (Gold 1978; Higuera 1997)

- $\mathbb{R}$ is a set of finite representations.
- For any $r \in \mathbb{R}$, $N(r) = h \in \mathbb{H}$ is the represented function.
- Given a finite sample of the function encoded by the representation, the learner must converge on a solution $\hat{r} \in \mathbb{R}$.
- If the function encoded by $\hat{r}$ matches but $\hat{r}$ does not match r, the result is *weak*, if $\hat{r}$ matches, the result is *strong* (Clark 2013)

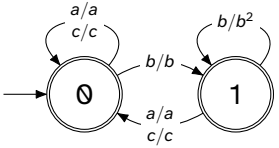
  Subsequential functions are identifiable in the limit from positive data in polynomial time and space (Oncina, Garcıa, and Vidal 1993).

Simple simplex systems

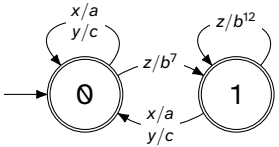
Simple simplex systems



(a) The homomorphism $r_\ell \in \mathbb{R}$.



(b) The simplex function $r_\rho \in \mathbb{R}$.



(c) The composition $r_\rho \circ r_\ell \in \mathbb{R}$.

Disclaimer

Notation alert

1. f_u: Tail function of string u on function f.
2. Tails(f): Set of tail functions for f.
3. $\mathring{f}_u(v)$: Common output of a string v on a tail f_u.
4. $\pi = (q, w, w', p)$: Path through a machine starting at a state q, reading a string w, outputting a string w', and ending at p.
5. $\omega^*(\pi) = w'$: Output of path π
6. $\iota^*(\pi) = w$: Input of path π
7. $\delta^*(\pi) = p$: Ending state of a path π

Homomorphic functions

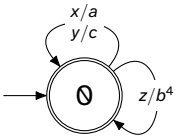


Figure: Machine of the homomorphism ℓ (r_ℓ).

Def. (Homomorphism)

A string-to-string function $f : A^* \to B^*$ is homomorphic iff for all $x, y \in A^*$, $f(xy) = f(x) \cdot f(y)$.

Left-simplex ISL_2 functions

c.f. Apx.

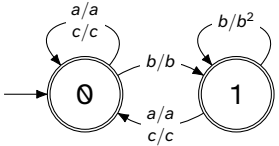


Figure: Machine of the left-simplex ISL_2 function ρ (r_ρ).

Def. ((left-)simplex ISL_2 function (Hua and Jardine 2021))

A properly ISL_2 function $\rho : B^* \to C^*$ is (left-)simplex iff there exists exactly one symbol $\sigma \in B$ and one tail function $\rho_{env} \in Tails(\rho)$ such that $\mathring{\rho}_{env}(\sigma) \neq \sigma$.

Learning perspective

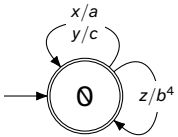
Core assumptions:

1. $\mathbb{H}_{\mathcal{H}\text{om}}$ is the space of homomorphisms.
2. $\mathbb{H}_{\text{ls-ISL}_2}$ is the space of left-simplex ISL_2 functions.
3. $\mathbb{R}_t$ is the space of *minimized, complete, and total canonical machines* for a given subclass t.

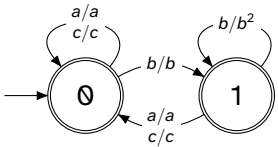
- The space SI_2DLA learns on is a subset of $\mathbb{R} = \mathbb{R}_{\text{ls-ISL}_2} \times \mathbb{R}_{\mathcal{H}\text{om}}$.
- SI_2DLA returns *at least* one composite representation $r_c \in \mathbb{R}$, and both functions $(h_\ell, h_\rho) \in \mathbb{H}$.

Coverage check

c.f. Apx.



(a) r_ℓ.

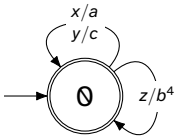


(b) r_ρ.

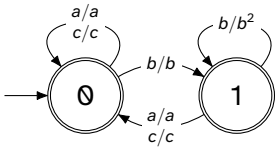
- The homomorphism is sufficient to uncover behavior of all ISL_2 tails. (witnessing)

Coverage check

c.f. Apx.



(a) r_ℓ.



(b) r_ρ.

- The homomorphism is sufficient to uncover behavior of all ISL_2 tails.
  (witnessing)
- The composition leaves enough information about the behavior of the simplex tails to identify them from the surface.
  (surface-witnessing)

Multiple solutions

Relevant xkcd

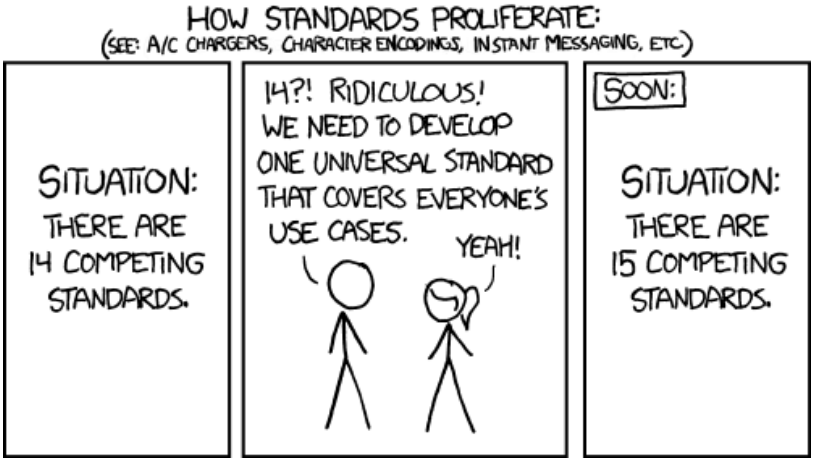


Figure: `https://xkcd.com/927`

Learning space

1. Fix $\mathbb{R}_c$ as some subclass of $\mathbb{R}$.
2. Assume an *a priori* order on all subclasses of $\mathbb{R}$.
3. For any ordering of subclasses, $\mathbb{R}_n \times \mathbb{R}_{n-1} \times \cdots \times \mathbb{R}_1 \subseteq \mathbb{R}^n$

   We write $\mathbb{R}^n$ in general
4. Let $\vec{r} = (r_n, r_{n-1}, \ldots, r_1) \in \mathbb{R}^n$
5. For any $\vec{r} \in \mathbb{R}^n$, $\vec{r}_\circ = r_n \circ r_{n-1} \circ \cdots \circ r_1$.

Learning space

1. Fix $\mathbb{R}_c$ as some subclass of $\mathbb{R}$.
2. Assume an *a priori* order on all subclasses of $\mathbb{R}$.
3. For any ordering of subclasses, $\mathbb{R}_n \times \mathbb{R}_{n-1} \times \cdots \times \mathbb{R}_1 \subseteq \mathbb{R}^n$

   We write $\mathbb{R}^n$ in general
4. Let $\vec{r} = (r_n, r_{n-1}, \ldots, r_1) \in \mathbb{R}^n$
5. For any $\vec{r} \in \mathbb{R}^n$, $\vec{r}_\circ = r_n \circ r_{n-1} \circ \cdots \circ r_1$.

Extend the same principles to $\mathbb{H}$.

Learner

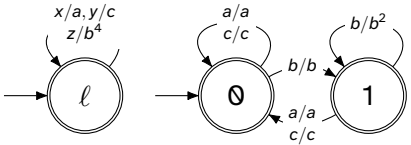
Def. ($(\mathbb{H}^n, \mathbb{R}^n)$-learning algorithm (Sketch))

A $(\mathbb{H}^n, \mathbb{R}^n)$-learning algorithm is a program $\mathcal{A}$ which takes a sample $D \subsetneq \vec{h}_\circ$ of a pair $(\vec{h}, \vec{r}) \in \mathbb{H}^n \times \mathbb{R}^n$ and returns a solution space $\mathcal{S} \subsetneq \mathfrak{P}(\mathbb{H}^n \times \mathbb{R}^n)$ such that for all $(\hat{\vec{h}}, \hat{\vec{r}}) \in \mathcal{S}$, $\hat{\vec{h}}_\circ = \vec{h}_\circ$ and $\hat{\vec{r}}_\circ = \vec{r}_\circ$.

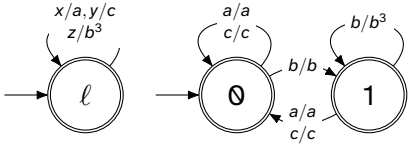
Pitch′

Compositional* equivalence of distinct solutions in shared classes.

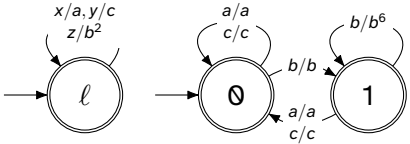
What does this look like?



(a) $\mathcal{S}_0$



(b) $\mathcal{S}_1$



(c) $\mathcal{S}_2$

We're looking at a $\mathbb{H}^2, \mathbb{R}^2$ learning problem, where $\mathbb{H}_2 = \mathbb{H}_{\text{ls-ISL}_2}$ and $\mathbb{H}_1 = \mathbb{H}_{\mathcal{H}\text{om}}$.

Applying the framework

c.f. Apx.

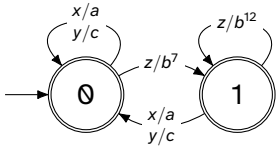


Figure: $r_\rho \circ r_\ell$.

1. Determine q_{env} and q_{def} (c.f. Hua and Jardine 2021).

Applying the framework

c.f. Apx.

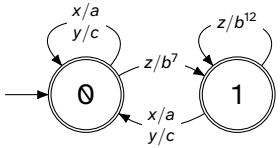


Figure: $r_\rho \circ r_\ell$.

1. Determine q_{env} and q_{def} (c.f. Hua and Jardine 2021).
   $q_{\text{env}} = 1$, $q_{\text{def}} = 0$

Applying the framework

c.f. Apx.

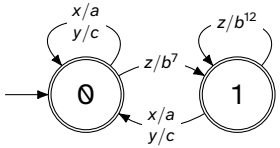


Figure: $r_\rho \circ r_\ell$.

1. Determine q_{env} and q_{def} (c.f. Hua and Jardine 2021).
   $q_{\text{env}} = 1$, $q_{\text{def}} = 0$
2. If $(q \neq q_{\text{env}}) \vee (b \neq \sigma)$, then $\omega(q, b) = b$ for all $q \in Q, b \in B$.

Applying the framework

c.f. Apx.

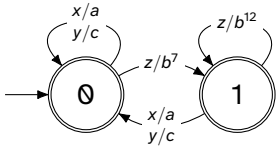
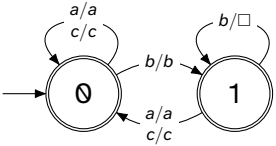


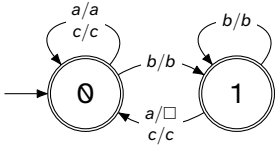
Figure: $r_\rho \circ r_\ell$.

1. Determine q_{env} and q_{def} (c.f. Hua and Jardine 2021).
   $q_{\text{env}} = 1$, $q_{\text{def}} = 0$
2. If $(q \neq q_{\text{env}}) \vee (b \neq \sigma)$, then $\omega(q, b) = b$ for all $q \in Q, b \in B$.
   We know q_{env}, we don't know σ

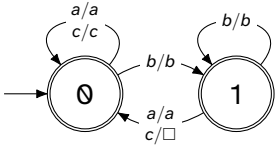
Simplex templates



(a) $r_{\hat{\rho}_0}$ $(b = \sigma)$.



(b) $r_{\hat{\rho}_1}$ $(a = \sigma)$.



(c) $r_{\hat{\rho}_2}$ $(c = \sigma)$.

Figure: Template machines.

Simplex templates: Evaluation

How do we evaluate templates?

Simplex templates: Evaluation

How do we evaluate templates? For any edge (q, a, w, p) in the composed machine, there is a path $(r, \ell(a), w, s) \in \Pi$ through the simplex machine, where $r = q \wedge s = p$.

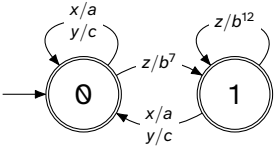
Simplex templates: Evaluation

How do we evaluate templates? For any edge (q, a, w, p) in the composed machine, there is a path $(r, \ell(a), w, s) \in \Pi$ through the simplex machine, where $r = q \wedge s = p$.

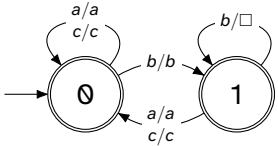
Insight

A successful hypothesis for a template will have exactly one path for each edge in the composed machine, and among all paths π, π' deriving $(q, a, w, p), (q', a', w', p')$, if $a = a'$ then $\iota^*(\pi) = \iota^*(\pi')$.

Simplex templates: Evaluation



(a) $r_\rho \circ r_\ell$.

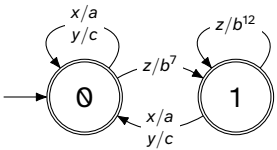


(b) Template $r_{\hat{\rho}_0}$.

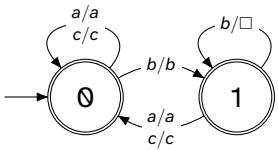
To evaluate:

$$\left\{ \begin{array}{lll} (0, x, a, 0), & (0, y, c, 0), & (0, z, b^7, 1), \\ (1, z, b^{12}, 1), & (1, x, a, 1), & (1, y, c, 1) \end{array} \right\}$$

Path tree description



(a) $r_\rho \circ r_\ell$.



(b) Template $r_{\hat{\rho}_\emptyset}$.

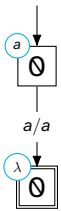
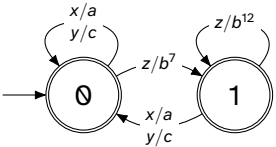


Figure: Path tree for $(\emptyset, x, a, \emptyset)$ on $r_{\hat{\rho}_\emptyset}$.

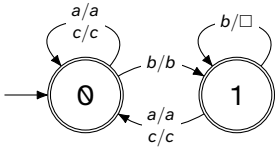
For any string (q, a, w, p) to derive:

1. Boxes are states in the path π (stages)
2. Cyan circled string is the remaining buffer at the stage $(\omega^*(\pi)^{-1} \cdot w)$
3. Transitions between stages are taken from the template machine
4. Doubled boxes are stages equal to p and for which $\omega^*(\pi) = w$

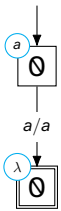
Path tree description



(a) $r_\rho \circ r_\ell$.



(b) Template $r_{\hat{\rho}_\emptyset}$.

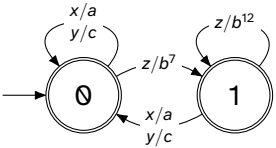


For any string (q, a, w, p) to derive:

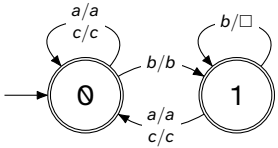
5. Buffer is calculated as the right quotient of the previous buffer w.r.t. output on the transition.
6. If the buffer is empty, then $\omega^*(\pi) = w$.

Figure: Path tree for $(\emptyset, x, a, \emptyset)$ on $r_{\hat{\rho}_\emptyset}$.

Building a path tree



(a) $r_\rho \circ r_\ell$.

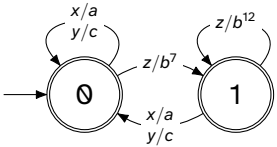


(b) Template $r_{\hat{\rho}_0}$.

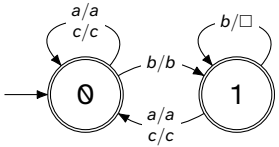
For any edge (r, b, u, s) to be added to a path π deriving (q, a, w, p):

- $\delta^*(\pi) = r$ (Contiguous)
- $\omega^*(\pi) \cdot u \in \Pr(w)$ (Contributing)
- $(r \neq s) \vee (u \neq \lambda)$ (Naıve anti-loop)

Decision conditions



(a) $r_\rho \circ r_\ell$.



(b) Template $r_{\hat{\rho}_\emptyset}$.

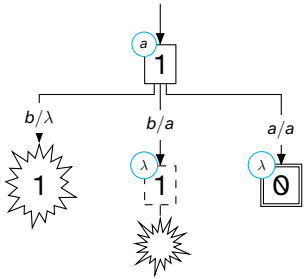


Figure: Path tree for $(1, x, a, \emptyset)$ on $r_{\hat{\rho}_\emptyset}$.

More path tree notation:

- Explosion with stage information: Violation of loop condition
- Explosion with no stage information: No compatible transition
- Dashed stage: New hypothesis (must remain internally consistent)

What counts as a hypothesis?

When a path reaches a blank output (□) in the template, we must make a hypothesis.

- Must not be the identity.
- Must be a prefix of buffered output.

Path tree derivation with multiple hypotheses

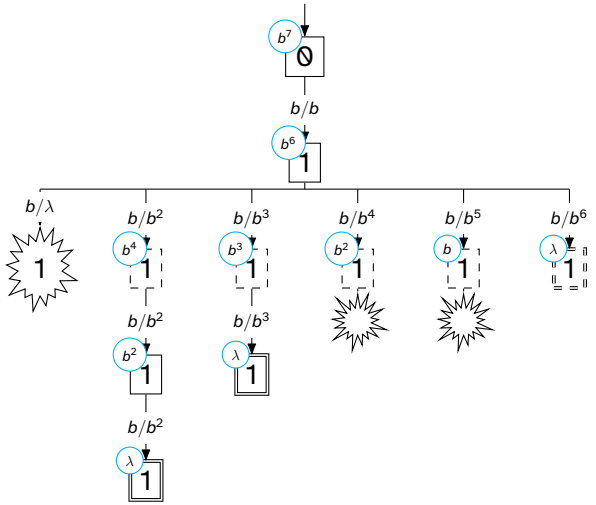


Figure: Path tree for $(\emptyset, z, b^7, 1)$ on $r_{\hat{\rho}_\emptyset}$.

Unifying a hypothesis

Insight (Restated)

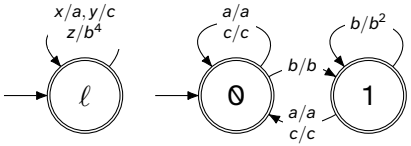
A successful hypothesis for a template will have exactly one path for each edge in the composed machine, and among all paths π, π' deriving $(q, a, w, p), (q', a', w', p')$, if $a = a'$ then $\iota^*(\pi) = \iota^*(\pi')$.

Valid paths which do not cross a hypothesis boundary will vacuously agree with all hypotheses.

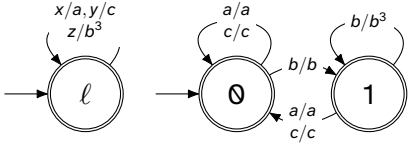
Unification

If each edge (q, a, w, p) has a single path π through a hypothesis machine such that a has a consistent hypothesis homomorphism, then the hypothesis holds.

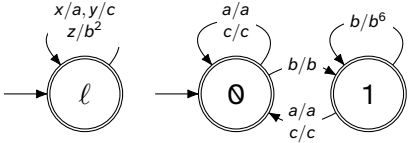
Compositionally equivalent solutions



(a) $\mathcal{S}_0$



(b) $\mathcal{S}_1$



(c) $\mathcal{S}_2$

Comparisons of solutions

Hypothetical measures

- $M_\rho = (k-1)+\mathrm{abs}(|\sigma|-|\mathring{\rho}_{\mathrm{env}}(\sigma)|)$
- $M_\ell = \sum_{w\in \mathrm{img}\ \ell} |w|$

	M_ρ	M_ℓ
$\mathcal{S}_0$	2	6
$\mathcal{S}_1$	3	5
$\mathcal{S}_2$	6	4

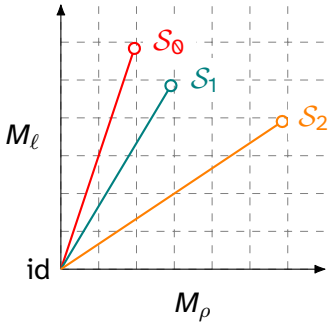


Figure: Graph of complexity measures.

Conclusion

This work:

- Diversity is a desirable trait[†]
- Meaningfully distinct solutions in shared complexity classes are possible for simple simplex systems

Further (current) work:

1. Applying this approach to more complicated simple systems (general simplex ISL_k)
2. Coverage from $\mathrm{SI}_2\mathrm{DLA}$.

References

- Clark, Alexander (2013). “Learning trees from strings: A strong learning algorithm for some context-free grammars”. In: *The Journal of Machine Learning Research* 14.1, pp. 3537–3559.
- Gold, E Mark (1978). “Complexity of automaton identification from given data”. In: *Information and control* 37.3, pp. 302–320.
- Higuera, Colin de la (1997). “Characteristic sets for polynomial grammatical inference”. In: *Machine Learning* 27.2, pp. 125–138.
- Hua, Wenyue and Adam Jardine (2021). “Learning input strictly local functions from their composition”. In: *International Conference on Grammatical Inference*. PMLR, pp. 47–65.
- Oncina, Jose, Pedro Garcıa, and Enrique Vidal (1993). “Learning subsequential transducers for pattern recognition interpretation tasks”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 15.5, pp. 448–458.

Appendix

Left-simplex ISL_2 functions

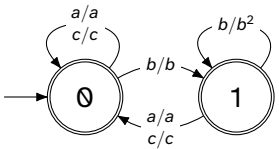


Figure: Machine of the left-simplex ISL_2 function ρ (r_ρ).

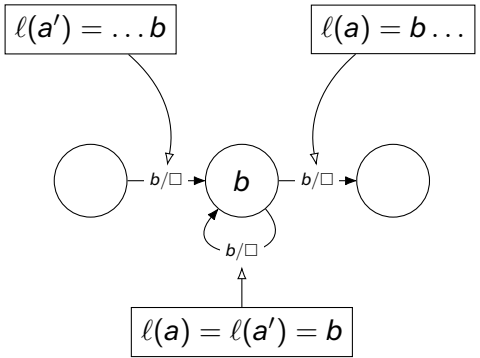
Def. ((left-)simplex ISL_2 function (Hua and Jardine 2021))

A properly ISL_2 function $\rho : B^* \to C^*$ is (left-)simplex iff there exists exactly one symbol $b \in B$ and one tail function $\rho_{env} \in Tails(\rho)$ such that $\mathring{\rho}_{env}(\rho) \neq \rho$.

$\rho_t \in \mathrm{Tails}(\rho)$	$b \in B$	$\mathring{\rho}_t(b)$	$\rho_t \in \mathrm{Tails}(\rho)$	$b \in B$	$\mathring{\rho}_t(b)$
ρ_{def}	a	a	ρ_{env}	a	a
	b	b		b	b^2
	c	c		c	c

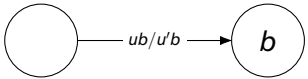
Def. (Witnessing ISL_2 (Hua and Jardine 2021))

A homomorphism $\ell : A^* \to B^*$ witnesses the ISL_2 class iff for all $b \in B$, there exists $a, a' \in A$ such that $Pr_1(\ell(a)) = b = Sf_1(\ell(a'))$.

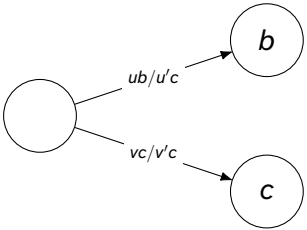


Def. (Opacity (Hua and Jardine 2021))

A string $w \in B^*$ is opaque with respect to a left-simplex ISL$_2$ function $\rho : B^* \to C^*$ iff $\rho_w \neq \rho_{w'}$, where $w' = \mathring{\rho}(w)$.



(a) Non-opaque string $w = ub$.



(b) Opaque string $w = ub$.

Surface-witnessing

Def. (Surface-witnessing (Hua and Jardine 2021))

A homomorphism $\ell : A^* \to B^*$ surface-witnesses a left-simplex ISL$_2$ function $\rho : B^* \to C^*$ iff there exists a subfunction $\ell' \subseteq \ell$ such that ℓ' witnesses ρ and there are no opaque strings in the image of ℓ'.

Show example is sufficient

Fix $A = \{x, y, z\}, B = \{a, b, c\}$.

1. Does ℓ witness the ISL_2 class?

Show example is sufficient

Fix $A = \{x, y, z\}, B = \{a, b, c\}$.

1. Does ℓ witness the ISL_2 class?

$b \in B$	$\sigma \in A$	$\mathrm{Pr}_1(\ell(\sigma))$	$\varsigma \in A$	$\mathrm{Sf}_1(\ell(\varsigma))$
a	x	a	x	a
b	z	b	z	b
c	y	c	y	c

Show example is sufficient

Fix $A = \{x, y, z\}, B = \{a, b, c\}$.

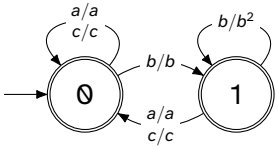
1. Does ℓ witness the ISL_2 class? ✓

Show example is sufficient

Fix $A = \{x, y, z\}, B = \{a, b, c\}$.

1. Does ℓ witness the ISL_2 class? ✓
2. Does ℓ surface-witness ρ?

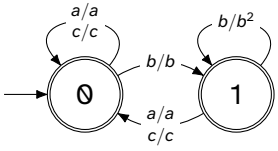
Show example is sufficient



Fix $A = \{x, y, z\}, B = \{a, b, c\}$.

1. Does ℓ witness the ISL_2 class? ✓
2. Does ℓ surface-witness ρ? We know $\ell \subseteq \ell$.

Show example is sufficient

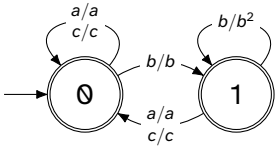


Fix $A = \{x, y, z\}, B = \{a, b, c\}$.

1. Does ℓ witness the ISL_2 class? ✓
2. Does ℓ surface-witness ρ? We know $\ell \subseteq \ell$.

$w \in \operatorname{img} \ell$	ρ_w	$\mathring{\rho}(w) = w'$	$\rho_{w'}$	Non-opaque?

Show example is sufficient

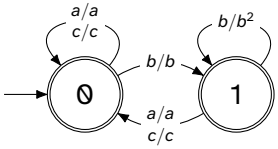


Fix $A = \{x, y, z\}, B = \{a, b, c\}$.

1. Does ℓ witness the ISL_2 class? ✓
2. Does ℓ surface-witness ρ? We know $\ell \subseteq \ell$.

$w \in \operatorname{img} \ell$	ρ_w	$\mathring{\rho}(w) = w'$	$\rho_{w'}$	Non-opaque?
a	ρ_a			

Show example is sufficient

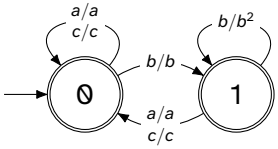


Fix $A = \{x, y, z\}, B = \{a, b, c\}$.

1. Does ℓ witness the ISL_2 class? ✓
2. Does ℓ surface-witness ρ? We know $\ell \subseteq \ell$.

$w \in \operatorname{img} \ell$	ρ_w	$\mathring{\rho}(w) = w'$	$\rho_{w'}$	Non-opaque?
a	ρ_a	a	ρ_a	

Show example is sufficient

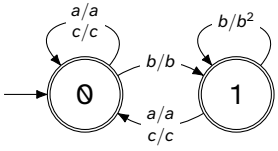


Fix $A = \{x, y, z\}, B = \{a, b, c\}$.

1. Does ℓ witness the ISL_2 class? ✓
2. Does ℓ surface-witness ρ? We know $\ell \subseteq \ell$.

$w \in \operatorname{img} \ell$	ρ_w	$\mathring{\rho}(w) = w'$	$\rho_{w'}$	Non-opaque?
a	ρ_a	a	ρ_a	✓

Show example is sufficient

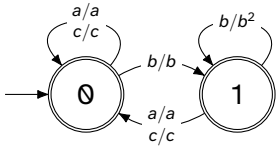


Fix $A = \{x, y, z\}, B = \{a, b, c\}$.

1. Does ℓ witness the ISL_2 class? ✓
2. Does ℓ surface-witness ρ? We know $\ell \subseteq \ell$.

$w \in \operatorname{img} \ell$	ρ_w	$\mathring{\rho}(w) = w'$	$\rho_{w'}$	Non-opaque?
a	ρ_a	a	ρ_a	✓
b^4	ρ_b	b^7	ρ_b	

Show example is sufficient

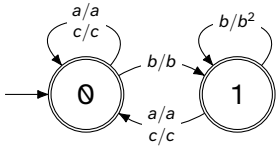


Fix $A = \{x, y, z\}, B = \{a, b, c\}$.

1. Does ℓ witness the ISL_2 class? ✓
2. Does ℓ surface-witness ρ? We know $\ell \subseteq \ell$.

$w \in \operatorname{img} \ell$	ρ_w	$\mathring{\rho}(w) = w'$	$\rho_{w'}$	Non-opaque?
a	ρ_a	a	ρ_a	✓
b^4	ρ_b	b^7	ρ_b	✓

Show example is sufficient

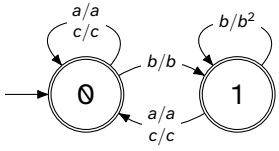


Fix $A = \{x, y, z\}, B = \{a, b, c\}$.

1. Does ℓ witness the ISL_2 class? ✓
2. Does ℓ surface-witness ρ? We know $\ell \subseteq \ell$.

$w \in \operatorname{img} \ell$	ρ_w	$\mathring{\rho}(w) = w'$	$\rho_{w'}$	Non-opaque?
a	ρ_a	a	ρ_a	✓
b^4	ρ_b	b^7	ρ_b	✓
c	ρ_c	c	ρ_c	

Show example is sufficient

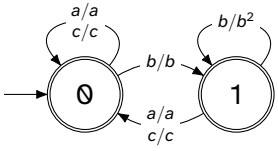


Fix $A = \{x, y, z\}, B = \{a, b, c\}$.

1. Does ℓ witness the ISL_2 class? ✓
2. Does ℓ surface-witness ρ? We know $\ell \subseteq \ell$.

$w \in \operatorname{img} \ell$	ρ_w	$\mathring{\rho}(w) = w'$	$\rho_{w'}$	Non-opaque?
a	ρ_a	a	ρ_a	✓
b^4	ρ_b	b^7	ρ_b	✓
c	ρ_c	c	ρ_c	✓

Show example is sufficient



Fix $A = \{x, y, z\}, B = \{a, b, c\}$.

1. Does ℓ witness the ISL_2 class? ✓
2. Does ℓ surface-witness ρ? ✓